

Run report

2026-10-04T022859Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	1	dataset	gaussian
seed	1,175,258,683	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	5.379×10^{-3}	best_epoch	98
_origin	{'batch_size': 'default', 'config': 'command-line', 'data_seed': 'default', 'dataset': 'default', 'hidden_units': 'default', 'lambda_dispersion': 'default', 'log_every': 'default', 'lr_d': 'default', 'lr_g': 'default', 'n_disc_steps': 'default', 'n_epochs': 'default', 'n_layers': 'default', 'n_samples': 'default', 'plots': 'default', 'run_dir': 'default', 'seed': 'default', 'variable': 'default'}	compile_cache_warm	True
gaussian_params	dim=1, mu_gen=[0.5], mu_true=[0], cov_gen=[[0.81]], cov_true=[[1]], cov_detector=[[0.25]]		
mmd_sigmas_detector	$0.7470 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$0.6688 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
data	0.051	1.0	gaussian cache hit – load
load	1.474	28.6	load
plots	3.627	70.3	load
evaluate	0.005	0.1	load
total	5.156	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	507.8	18.41	96.4	13.13	97.4	97.77	80.7

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	28.91	0.5955	97.9	0.7157	97.5	1.792	93.8

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	111.5	2.165	98.1	2.624	97.6	6.878	93.8

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	455.0	16.49	96.4	11.76	97.4	87.60	80.7

Mean 1D Wasserstein distance over random directions, in units of Data’s per-observable standard deviation. IBU is scored on the product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	508.5	42.23	91.7	12.35	97.6	97.44	80.8

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	37.03	7.828	78.9	0.7511	98.0	2.432	93.4

VLC divergence [$\times 10^3$]

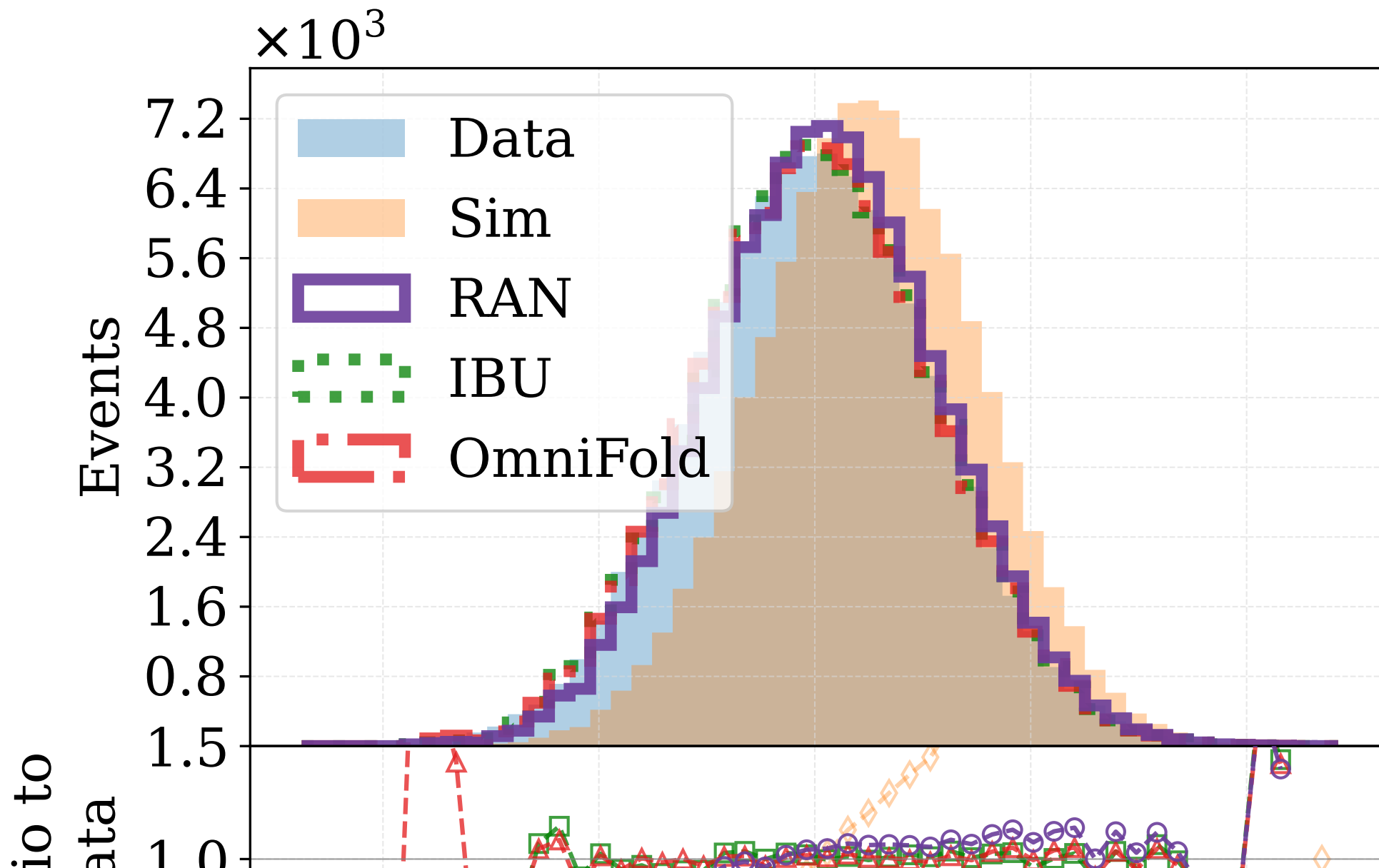
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	141.3	30.88	78.2	2.778	98.0	9.347	93.4

Sliced Wasserstein, joint [$\times 10^3$]

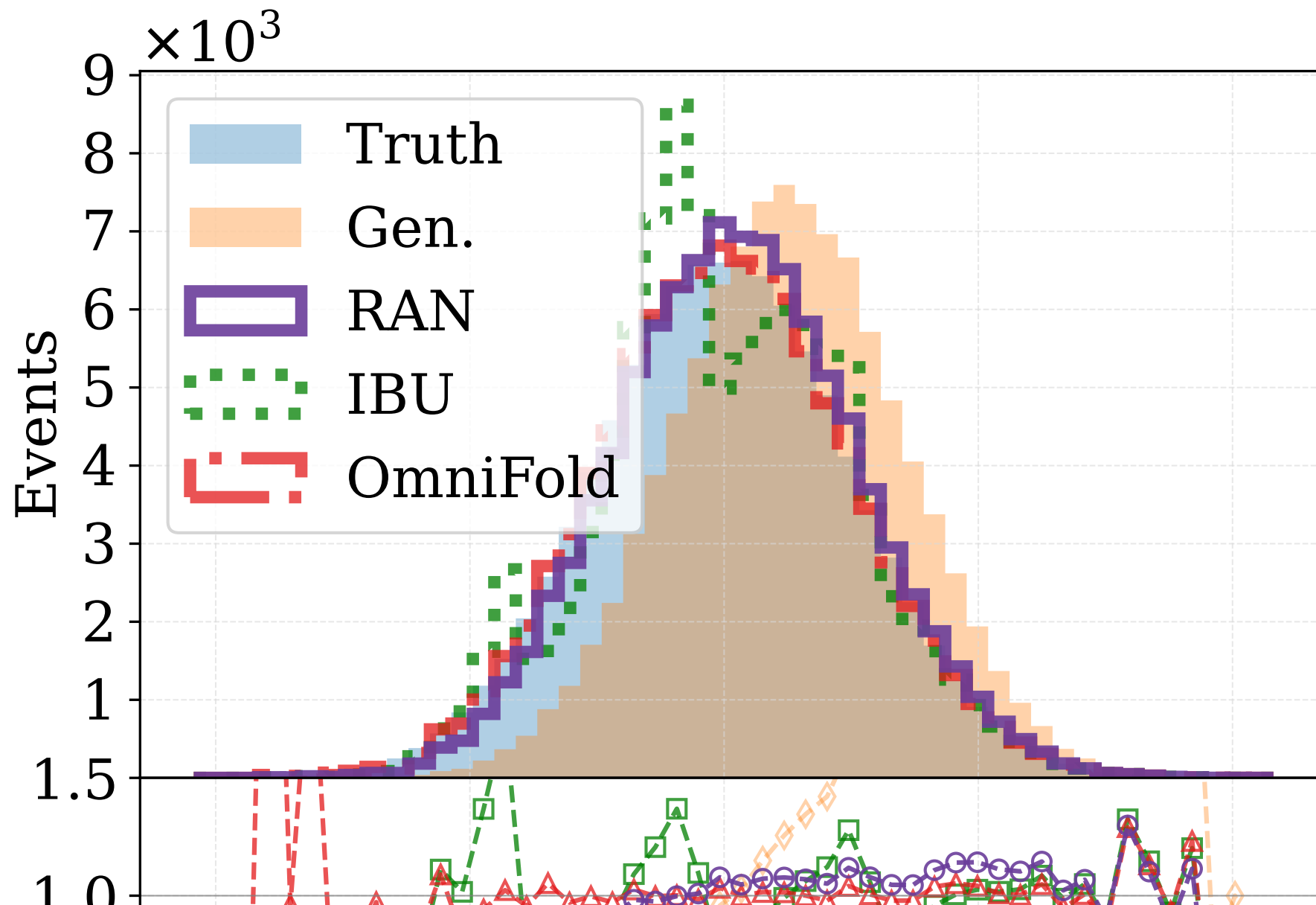
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
All	508.4	42.23	91.7	12.34	97.6	97.44	80.8

Mean 1D Wasserstein distance over random directions, in units of Truth’s per-observable standard deviation. IBU is scored on the product of its per-observable weights.

Detector Level

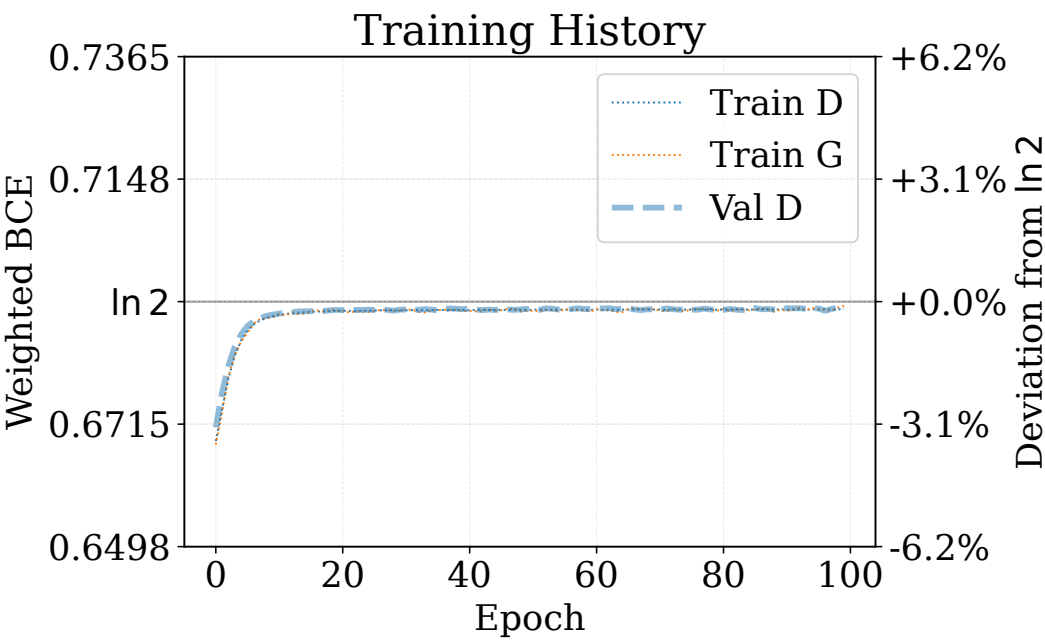


Particle Level



Training diagnostics

Training losses



Selection criteria

